DATE / NO. | TITLE

Problem Set 2 - Question 1

1a) Yes.

define $X$ as payoff matrix $X = \begin{pmatrix} 10 & 10 & 0 \\ 20 & 10 & 10 \\ 30 & 10 & 10 \end{pmatrix}$

where each column is payoff of one assee across different states

Complete markets $\Leftrightarrow$ we can find a unique matrix $\Omega$ s.t. $X\Omega = I$. Now find $\Omega$:

$$\begin{pmatrix} 10 & 10 & 0 \\ 20 & 10 & 10 \\ 30 & 10 & 10 \end{pmatrix}\begin{pmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

state 1:
$$\left.\begin{aligned} 10w_{11} + 10w_{22} &= 1 \\ 20w_{11} + 10w_{21} + 10w_{31} &= 0 \\ 30w_{11} + 10w_{21} + 10w_{31} &= 0 \end{aligned}\right\} \Rightarrow \begin{aligned} w_{11} &= 0 \\ w_{21} &= \frac{1}{10} \\ w_{31} &= -\frac{1}{10} \end{aligned}$$

state 2:
$$\left.\begin{aligned} 10w_{12} + 10w_{22} &= 0 \\ 20w_{12} + 10w_{22} + 10w_{32} &= 1 \\ 30w_{12} + 10w_{22} + 10w_{32} &= 0 \end{aligned}\right\} \Rightarrow \begin{aligned} w_{12} &= -\frac{1}{10} \\ w_{22} &= \frac{1}{10} \\ w_{32} &= \frac{1}{5} \end{aligned}$$

state 3:
$$\left.\begin{aligned} 10w_{13} + 10w_{23} &= 0 \\ 20w_{13} + 10w_{23} + 10w_{33} &= 0 \\ 30w_{13} + 10w_{23} + 10w_{33} &= 1 \end{aligned}\right\} \Rightarrow \begin{aligned} w_{13} &= \frac{1}{10} \\ w_{23} &= -\frac{1}{10} \\ w_{33} &= -\frac{1}{10} \end{aligned}$$

$\Rightarrow \Omega = \begin{pmatrix} 0 & -\frac{1}{10} & \frac{1}{10} \\ \frac{1}{10} & \frac{1}{10} & -\frac{1}{10} \\ -\frac{1}{10} & \frac{1}{5} & -\frac{1}{10} \end{pmatrix}$ is unique

$X\Omega = I$

So we have complete markets

DATE/NO. | TITLE

1.b) Note that $I$ is the payoff matrix of 3 contigent claims since $X\Pi = I$, by LOP,

$$P_{C}_{3\times 1} = \underset{3\times 3}{\Pi'} \underset{3\times 1}{P}$$

$$= \begin{pmatrix} w_{11} & w_{21} & w_{31} \\ w_{12} & w_{22} & w_{32} \\ w_{13} & w_{23} & w_{33} \end{pmatrix} \begin{pmatrix} 16 \\ 8 \\ 6 \end{pmatrix}$$

$$= \begin{pmatrix} 0\times 16 + \frac{1}{10}\times 8 + (-\frac{1}{10})\times 6 \\ -\frac{1}{10}\times 16 + \frac{1}{10}\times 8 + \frac{1}{5}\times 6 \\ \frac{1}{10}\times 16 + -\frac{1}{10}\times 8 + (-\frac{1}{10})\times 6 \end{pmatrix} = \begin{pmatrix} \frac{1}{5} \\ \frac{2}{5} \\ \frac{1}{5} \end{pmatrix}$$

(c) payoff: $X_D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 30 \\ 20 \\ 10 \end{pmatrix}$

by LOP: $P_D = P_C' \begin{pmatrix} 30 \\ 20 \\ 10 \end{pmatrix} = \frac{1}{5}\times 30 + \frac{2}{5}\times 20 + \frac{1}{5}\times 10 = 16.$

(d) an SDF must price all assees including Arrow-Debreu Claim

$P_t = E_t[M \cdot X_{t+1}]$

$P_C(n) = \pi(n) \cdot M(n) \cdot 1$

$\Rightarrow M(n) = \frac{P_C(n)}{\pi(n)}$

$$M = \begin{pmatrix} \frac{1}{5} / \frac{1}{4} \\ \frac{2}{5} / \frac{1}{2} \\ \frac{1}{5} / \frac{1}{4} \end{pmatrix} = \begin{pmatrix} \frac{4}{5} \\ \frac{4}{5} \\ \frac{4}{5} \end{pmatrix}$$

- $M(n) > 0$ for every state.
  suppose payoff $X_A \geq 0$. We will have $P_A = E[M \cdot X_A] \geq 0$.
  $\Rightarrow$ no arbierage.
- SDF in this economy is constant across states
  There is no state-dependent risk premium coming from SDF since $Cov_t(M, X_{t+1}) = 0$. A dollar is state 1 is valued the same as a dollar in state 2 or 3.

DATE / NO. | TITLE

(e) $P_{D,t} = E_t[M \cdot X_{D,t+1}]$

$= 0.8\left(\frac{1}{4} \cdot 30 + \frac{1}{2} \cdot 20 + \frac{1}{4} \cdot 10\right)$

$= 16$

is the same as calculated in (c)

DATE/NO. | TITLE

Q3 (a) $\mathcal{L} = E_t[U(W_t \cdot w_t' R_{t+1})] + \lambda_t(1 - 1'w_t)$

F.O.C. w.r.t $w_t$

$$\frac{\partial \mathcal{L}}{\partial w_t} = E_t[U'(W_{t+1}) W_t R_{t+1}] - \lambda_t \cdot 1 = 0$$

for any two assets $j, k$,

$$E_t[U'(W_{t+1}) W_t R_{j,t+1}] = E_t[U'(W_{t+1}) W_t R_{k,t+1}]$$

$$\Rightarrow E_t[U'(W_{t+1}) W_t (\underbrace{R_{j,t+1} - R_{k,t+1}}_{= \times R_{j,t+1} \text{ if } k \text{ is baseline asset } R_{f,t+1}})] = 0$$

Under CRRA, $U'(W_{t+1}) = W_{t+1}^{-\gamma}$

$$= W_t^{-\gamma} R_{w,t+1}^{-\gamma}$$

$$E_t[W_t^{-\gamma+1} R_{w,t+1}^{-\gamma} \times R_{j,t+1}] = 0$$

$$W_t^{-\gamma+1} E_t[R_{w,t+1}^{-\gamma} \times R_{j,t+1}] = 0$$

$$E_t[R_{w,t+1}^{-\gamma} \times R_{j,t+1}] = 0 \quad \text{for every } j$$

So $E_t[R_{w,t+1}^{-\gamma} \times R_{t+1}] = 0$

$\therefore$ the SDF $M_t = R_{w,t+1}^{-\gamma}$ prices $\times R_{t+1}$.

Q4 (a) Let $\mu_t = E_t[R_{w,t+1}]$

Using a first order Taylor expansion of $R_{w,t+1}^{-\gamma}$ around $\mu_t$,

$$M_{t+1} = R_{w,t+1}^{-\gamma}$$

$$\approx \mu_t^{-\gamma} - \gamma \mu_t^{-\gamma-1}(R_{w,t+1} - \mu_t)$$

$$= \underbrace{(1+\gamma)\mu_t^{-\gamma}}_{a_t} - \underbrace{\gamma \mu_t^{-\gamma-1}}_{b_t} \cdot R_{w,t+1}$$

disadvantage, linear SDF can be negative for very large $R_{w,t+1}$, while the original power-utility SDF $M_t = R_{w,t+1}^{-\gamma}$ is always positive as long as $R_{w,t+1} > 0$